

Research Seminar

Degradation analysis in Si-based lithium-ion batteries

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HANYANG UNIVERSITY

Part 1

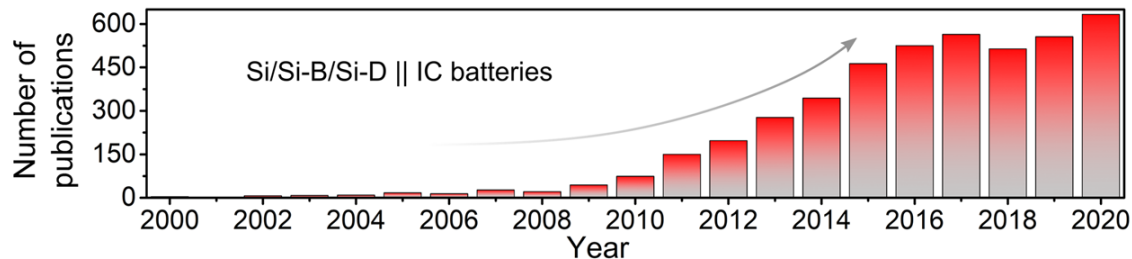
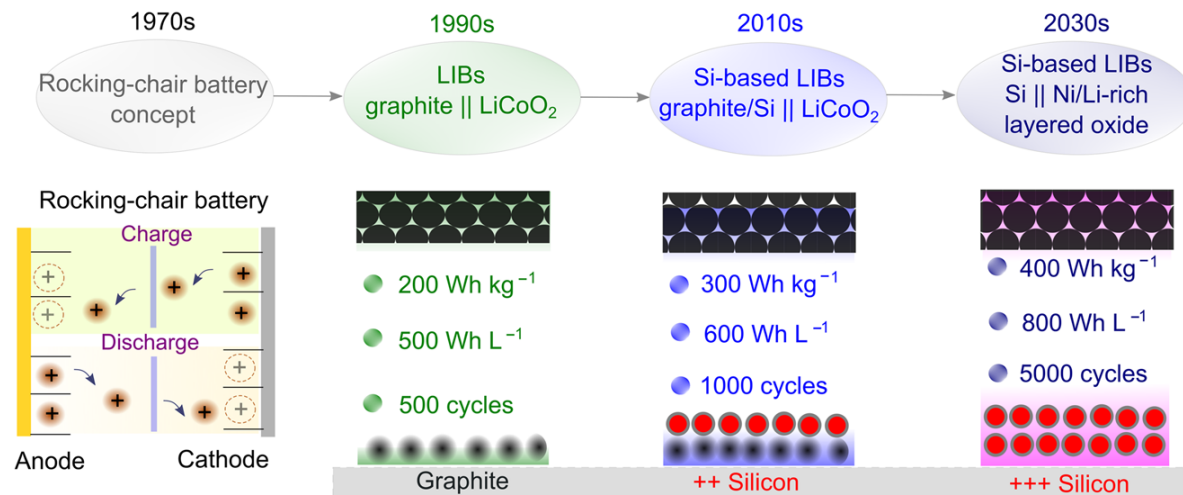
Origin of nonlinear state-of-health evolution
in high-Ni NCM || Si-graphite batteries

Introduction

■ Advantages of Si-graphite anode

- Graphite anodes have reached their theoretical limit in terms of energy density.
- The incorporation of Si into graphite anodes improves the energy density of lithium-ion batteries (LIBs).

Nature Communications, 2021, 12, 5459

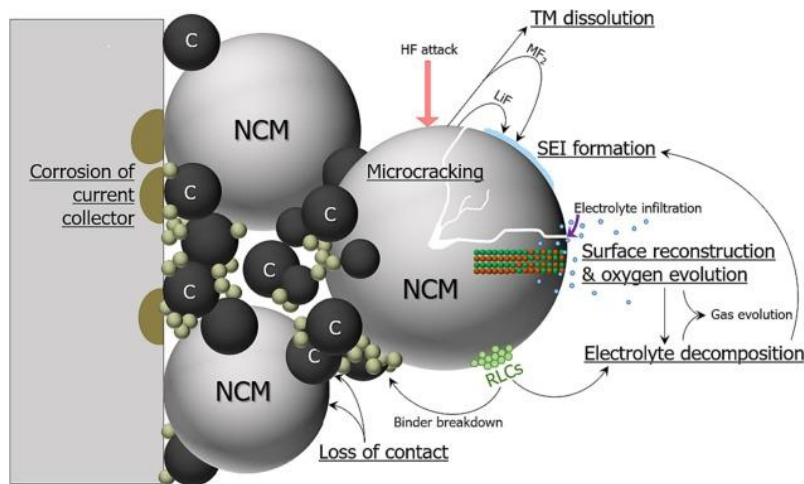


Introduction

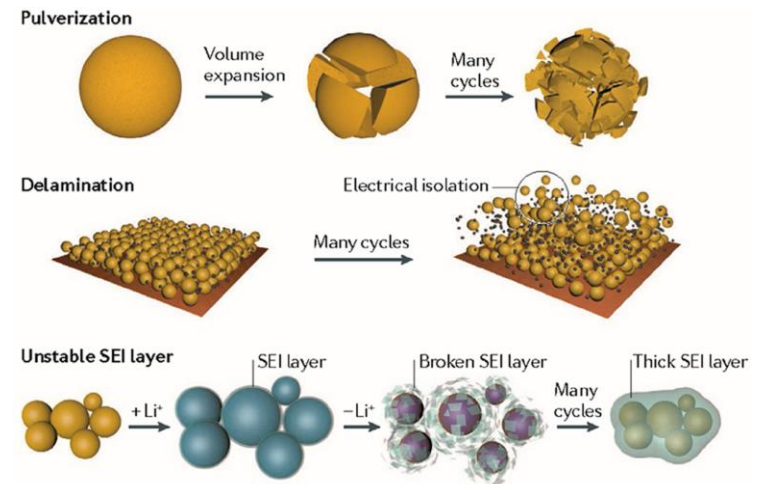
■ Degradation of NCM || Si-graphite cell

- Various degradation mechanisms affect to the cell configurations during the cycling.
- Distinguishing between these various degradation mechanisms is an important point of degradation analysis.

Journal of Energy Storage, 2023, 73, 108875

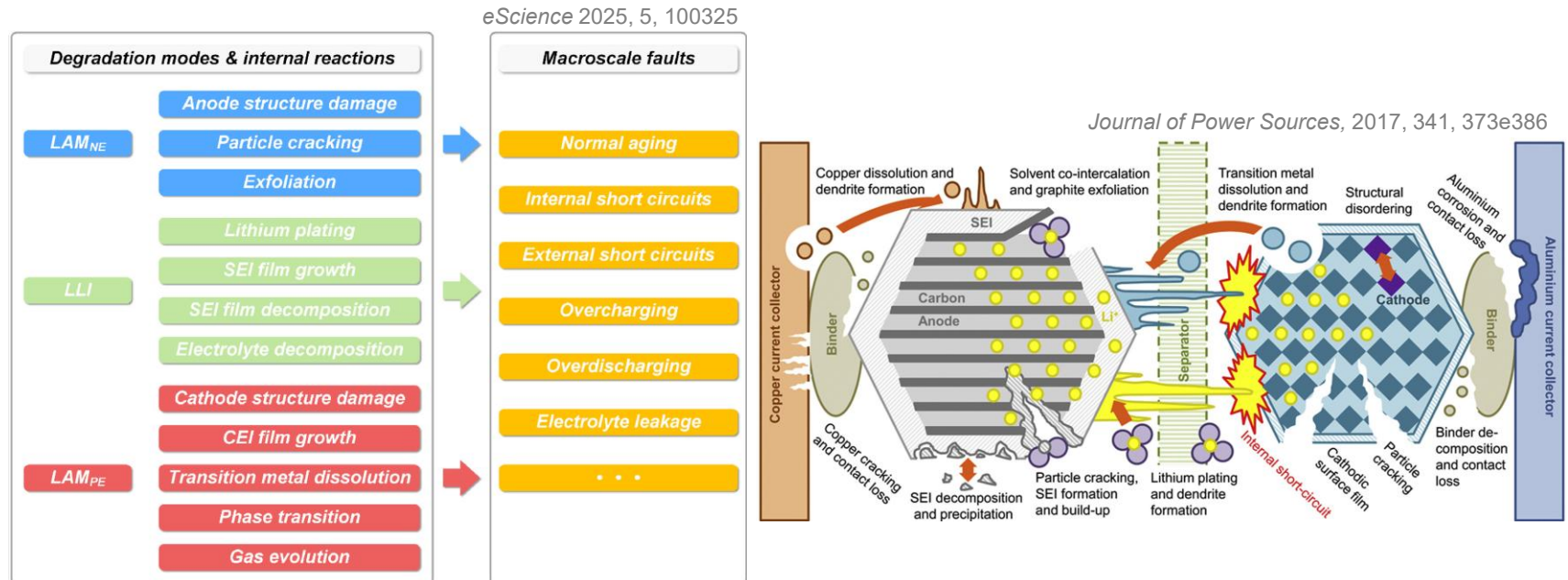


Energy Fuels, 2021, 35, 944–964



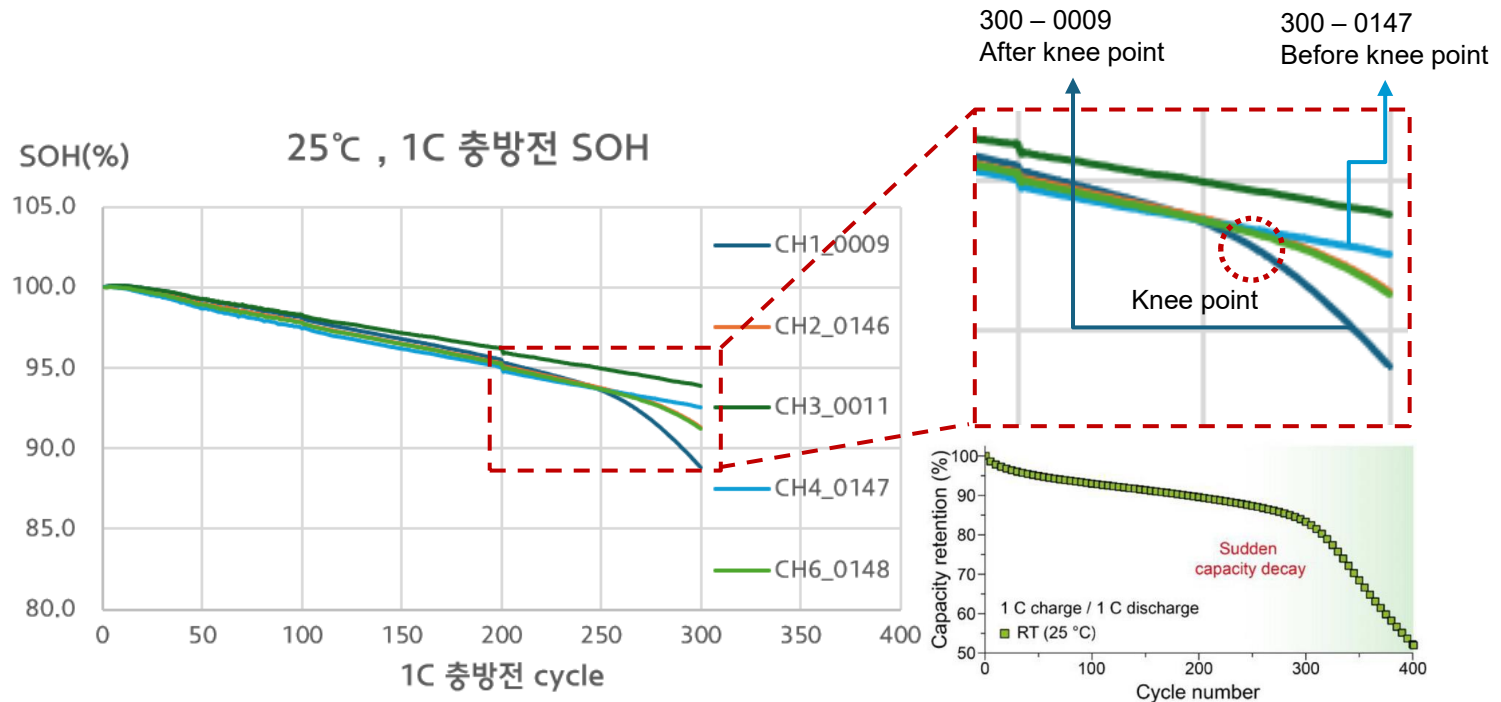
Introduction

- Importance of quantifying degradation mode
 - All macro-scale degradations are derived from the combination of degradation modes.
 - Quantifying degradation modes can predict macro-scale degradations and prevents prediction error.



Introduction

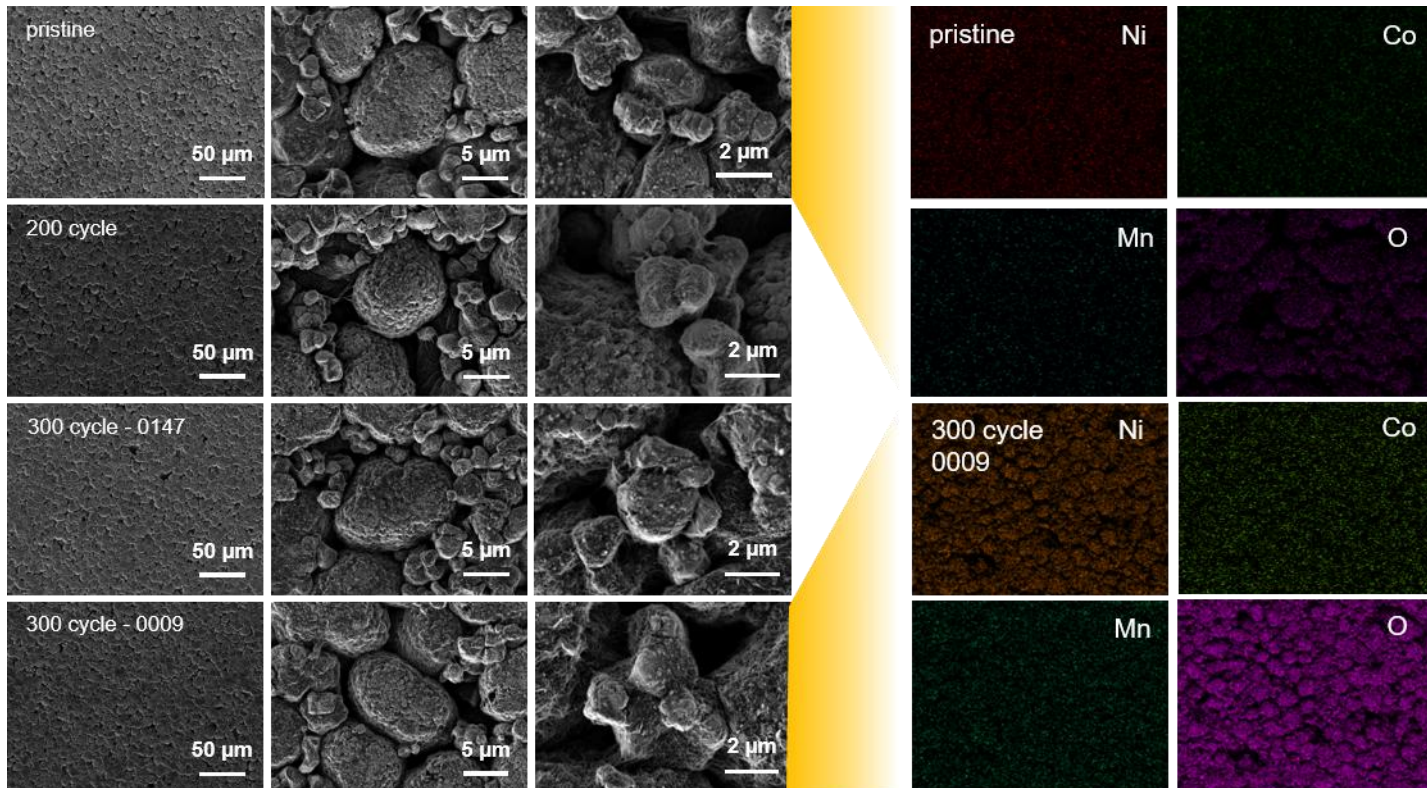
- Knee point identified in NCM || Si-graphite cell
 - Sudden **capacity decay** (knee point) about 300 cycle occurs in NCM || Si-graphite cell.
 - Analytical techniques for the non-destructive diagnosis of degradation mechanism are required.



Journal of Power Sources 2022, 532, 231296

Results & Discussion

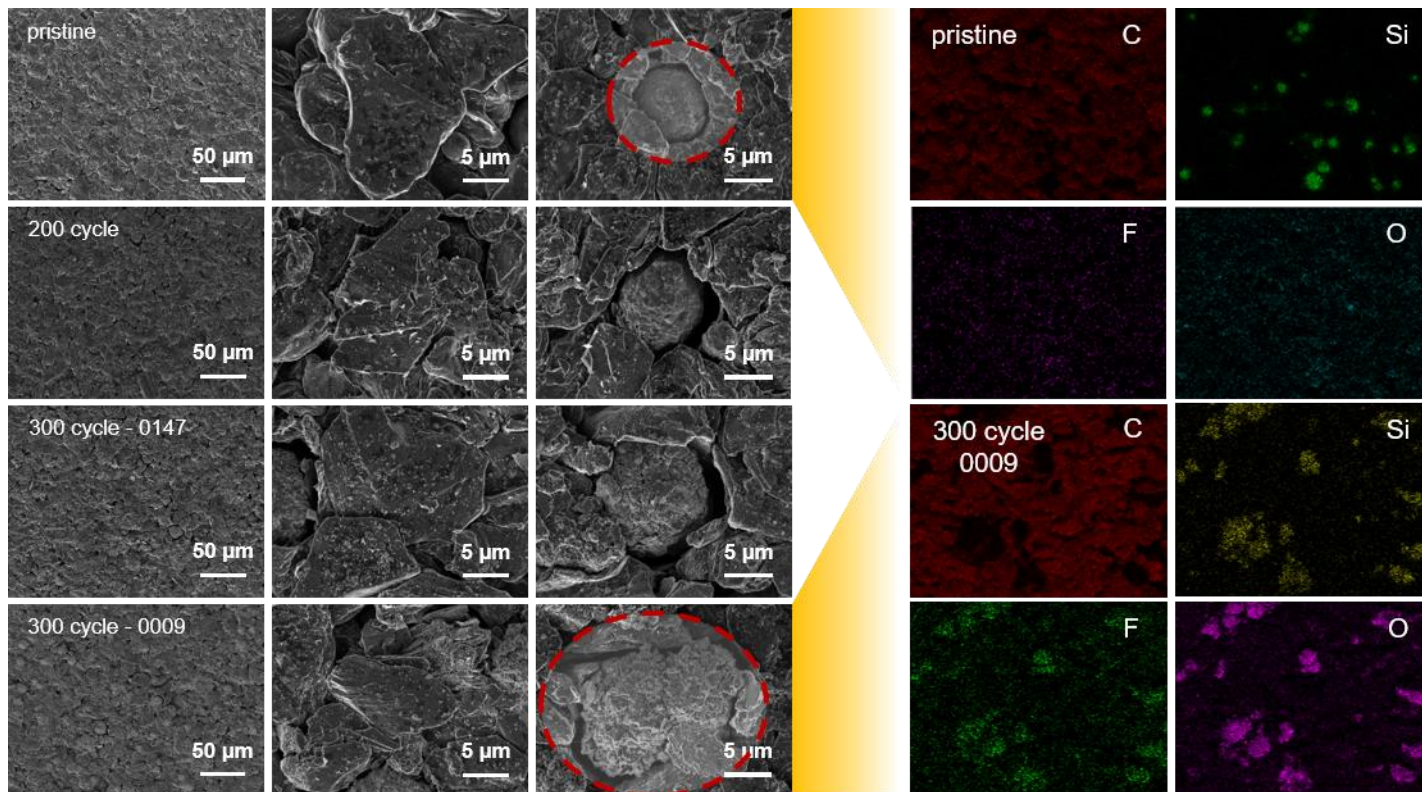
- SEM analysis for each cycle electrodes
 - The SEM images of the positive electrode (PE) showed no noticeable degradation.



Results & Discussion

■ SEM analysis for each cycle electrodes

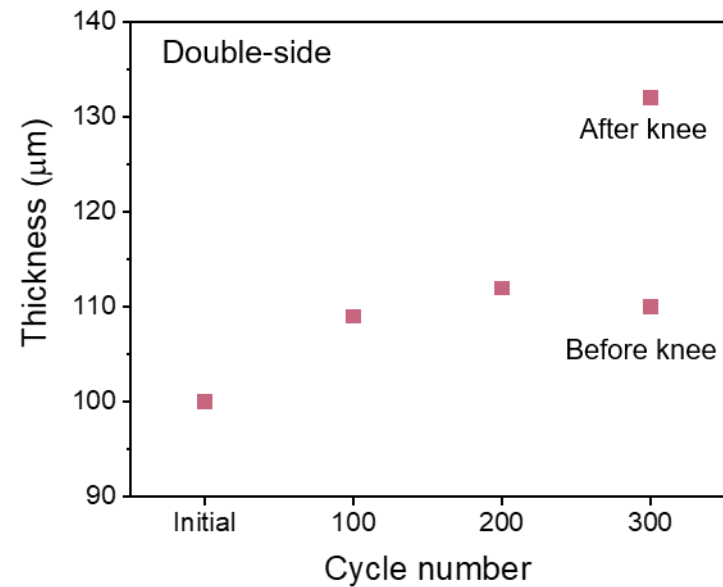
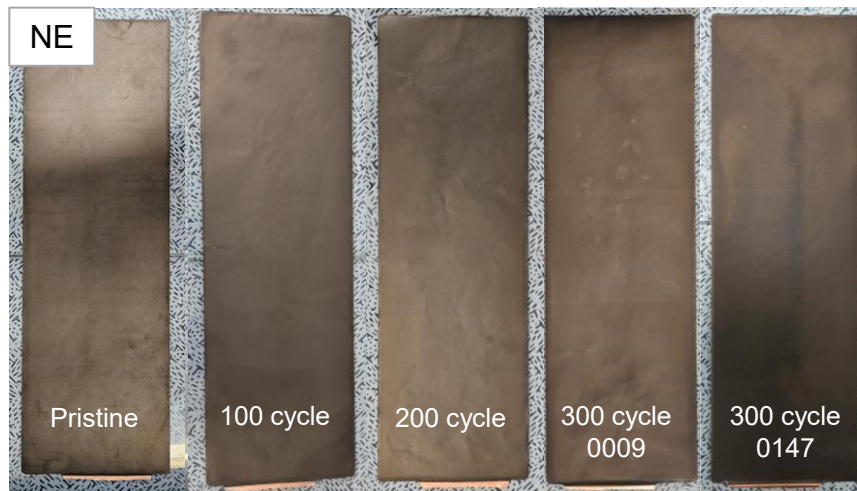
- The SiC particle of after-knee negative electrode (NE) lost their spherical shape and degraded due to volume expansion.
- O and F located in the same place with Si after degradation in EDS.



Results & Discussion

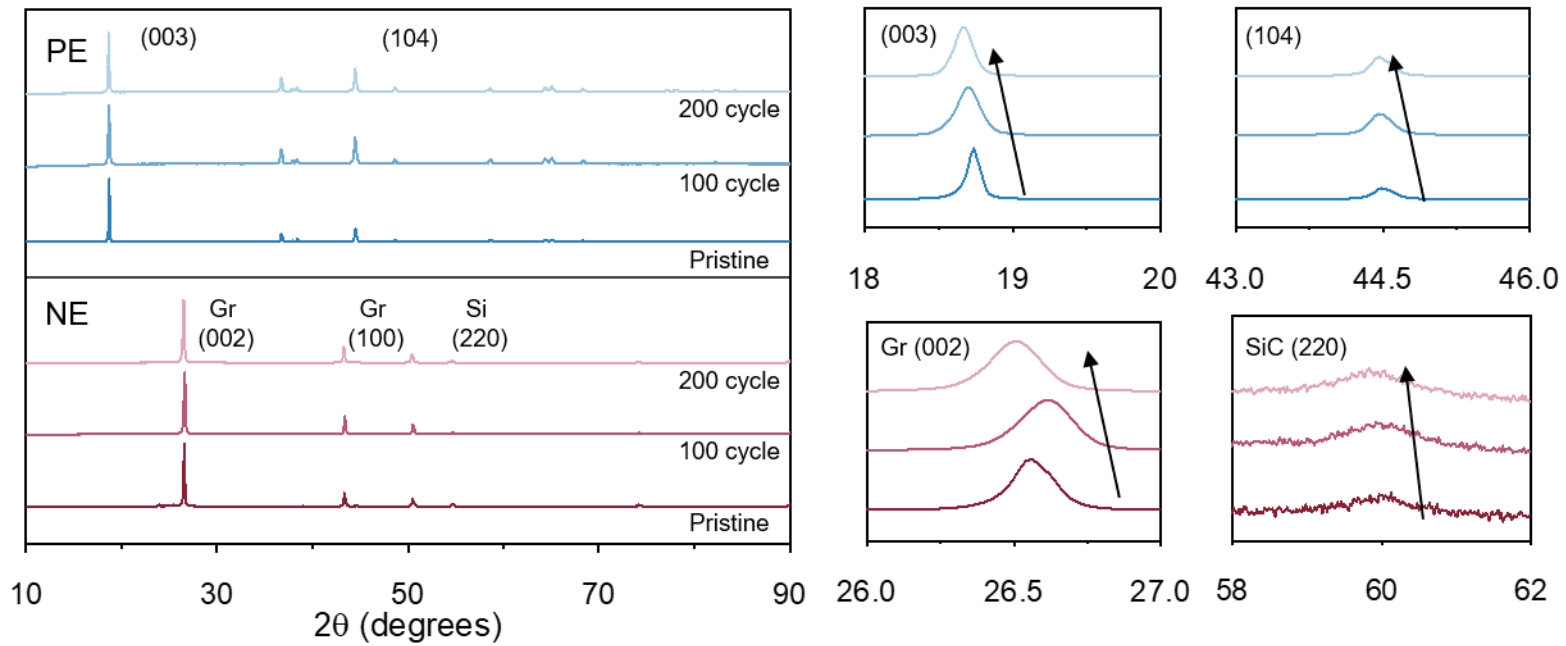
■ Electrode thickness measurement

- The thickness of the after-knee NE increased significantly due to volume expansion of the SiC particles.
- In contrast, before-knee NE at 300 cycle showed no thickness increment.



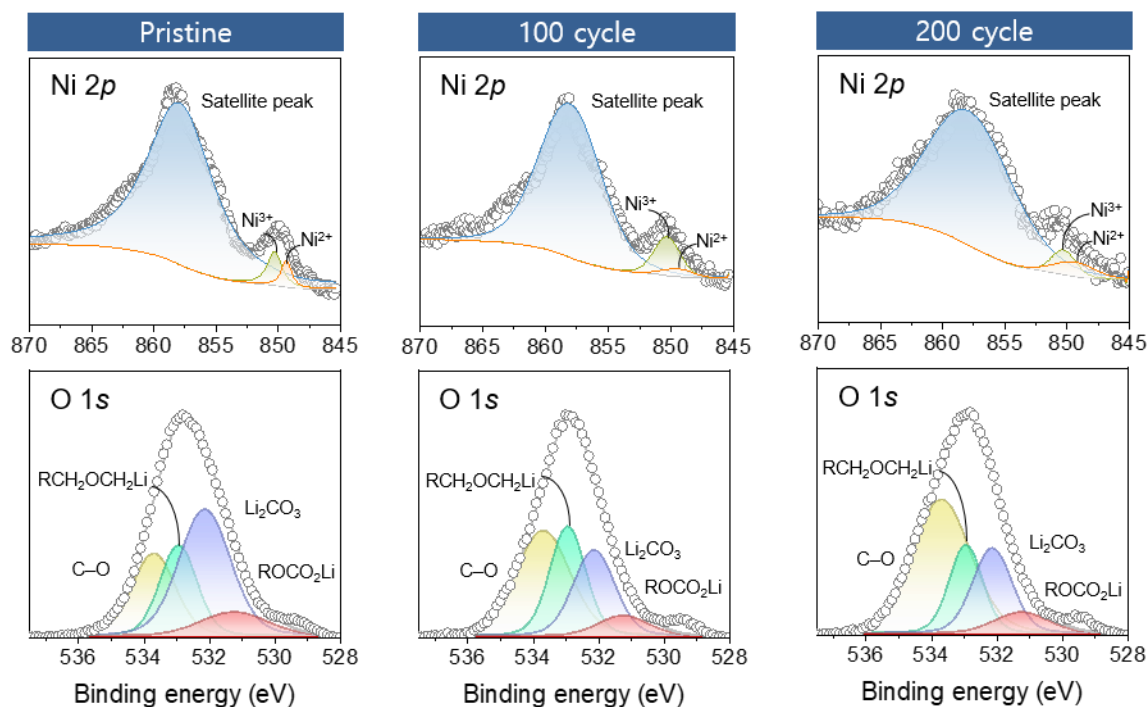
Results & Discussion

- XRD analysis for each cycle electrodes
 - The peaks were observed to progressively shift to the left with cycling.
 - The lattice structure of the active material irreversibly degraded.



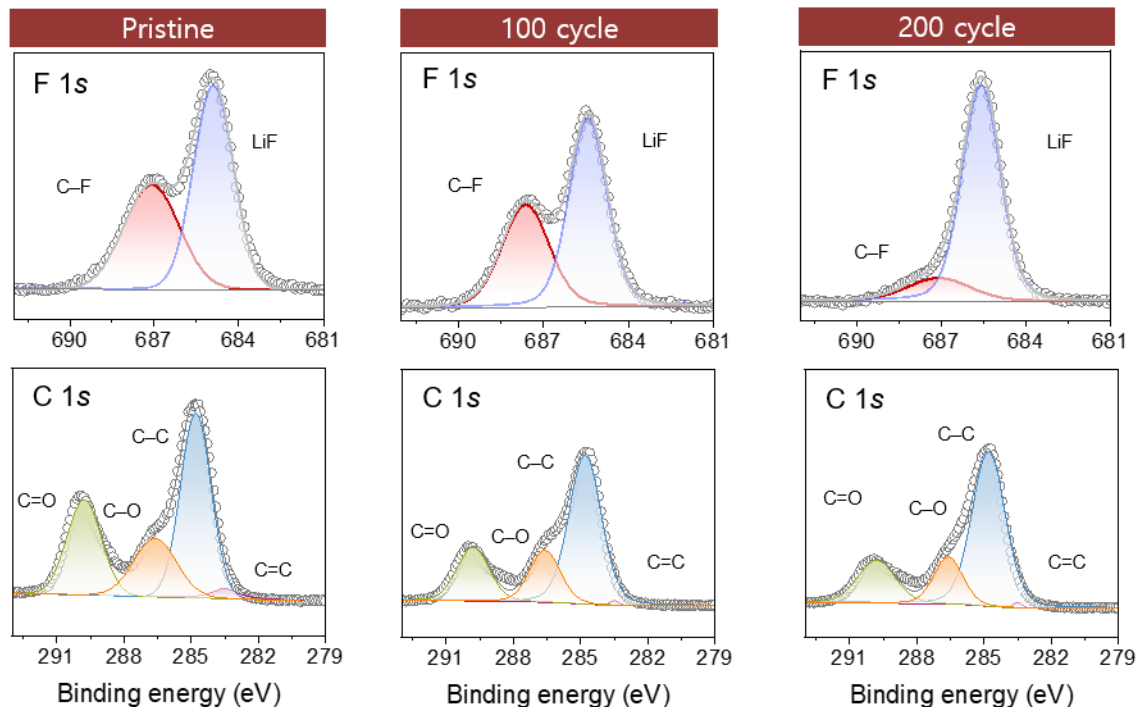
Results & Discussion

- XPS analysis for each cycle electrodes for PE
 - $\text{Ni}^{2+}/\text{Ni}^{3+}$ peak area ratio increases with cycling for PE.
 - PE degradation due to rock-salt phase transition on the active material.



Results & Discussion

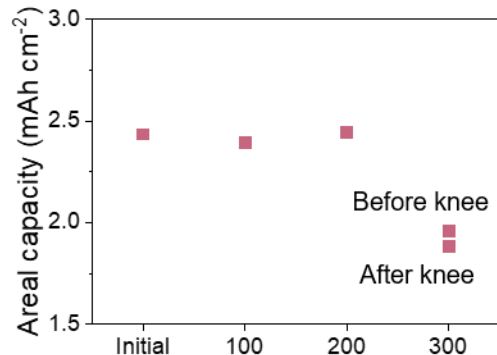
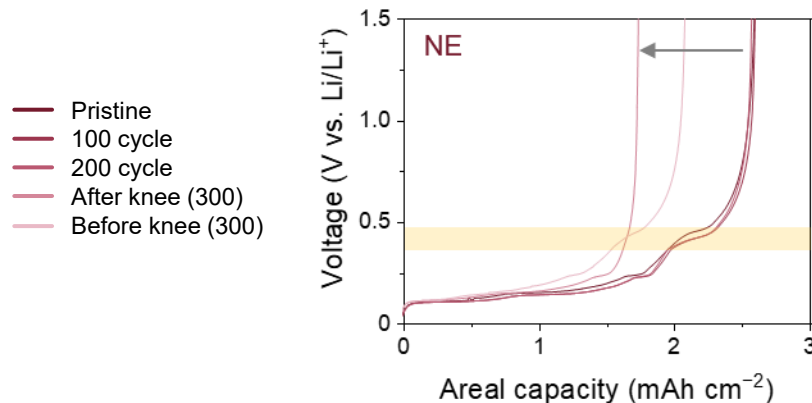
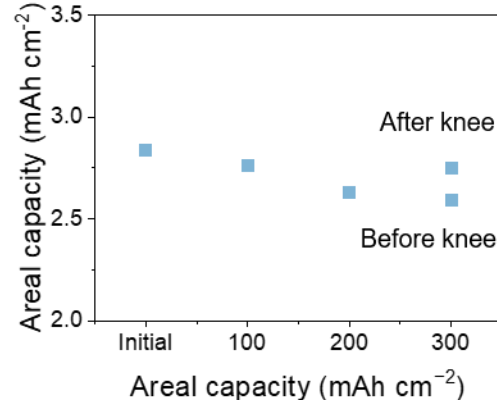
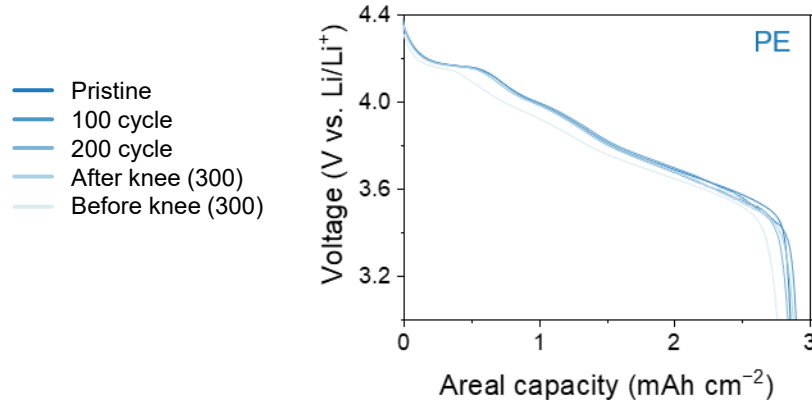
- XPS analysis for each cycle electrodes for NE
 - SEI-related peak intensity increases with cycling for NE.
 - The after-knee NE is expected to show a greater increase in intensity.



Results & Discussion

■ Half-cell formation cycling for each cycle

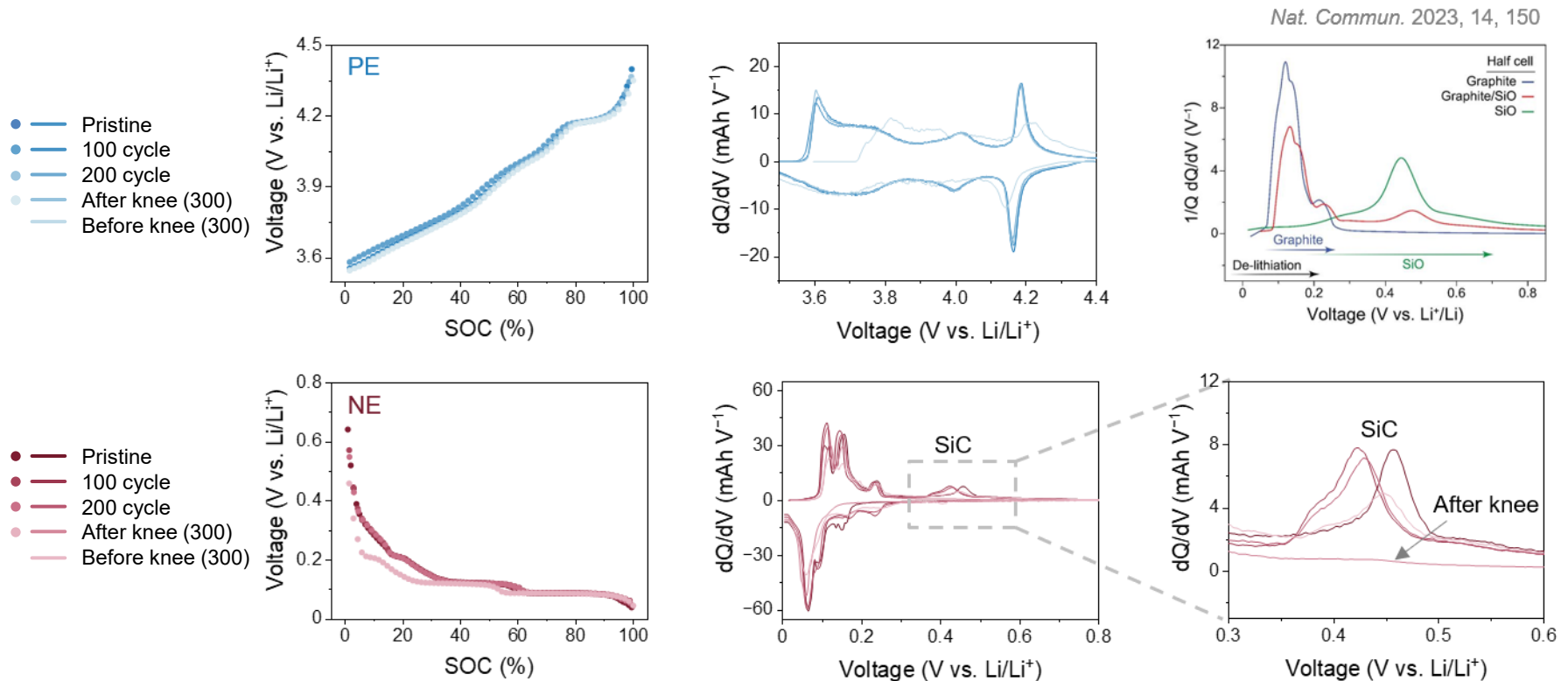
- Half-cells were cycled in formation condition with low C-rate to export capacity and pseudo-open circuit voltage (pOCV) for each cycled electrodes. (0.05Cx1 + 0.1Cx1)
- The 300cycle NE half cell showed noticeable capacity drop.



Results & Discussion

■ GITT and ICA evolution

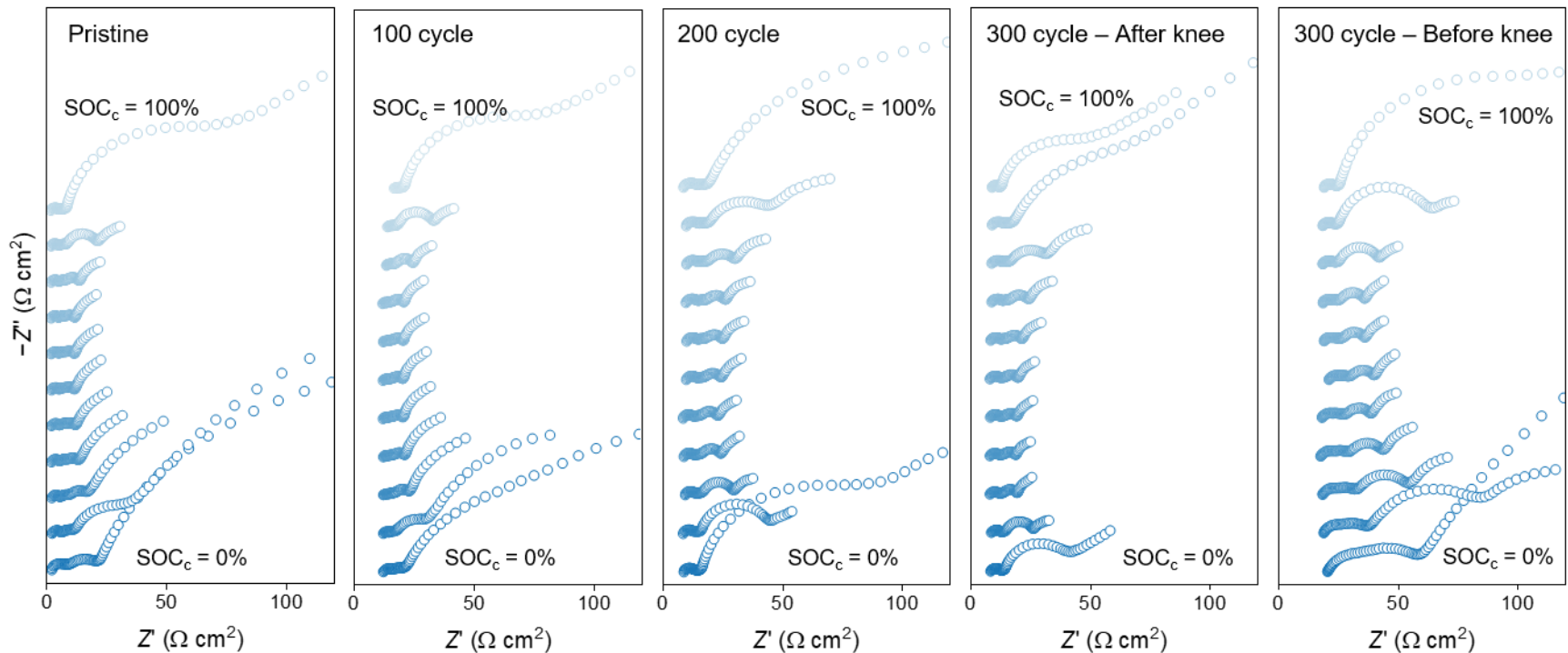
- The GITT-OCV at after-knee NE showed a voltage drop compared with other cycles.
- The SiC peak at ~0.4 V disappeared in the after-knee NE ICA curve.
- Before-knee electrodes showed high overpotential, especially in PE.



Results & Discussion

■ EIS evolution

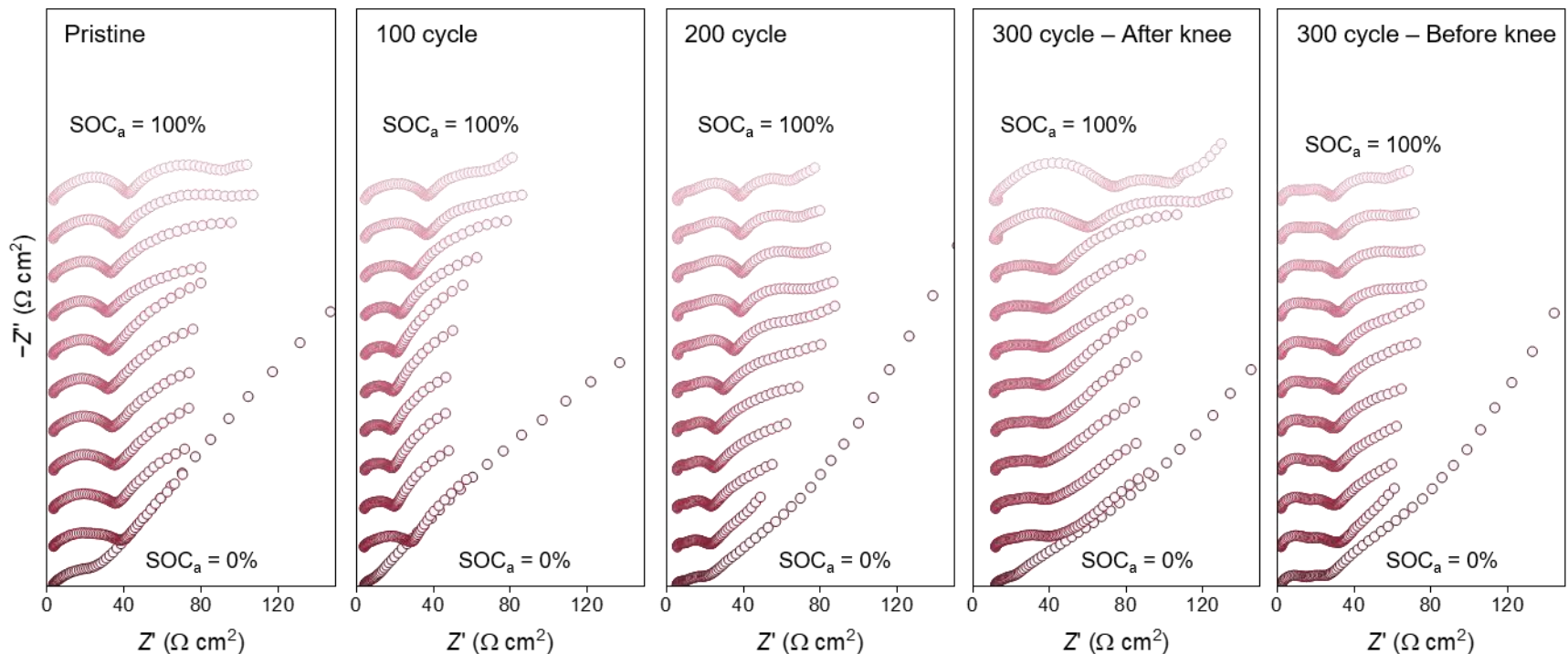
- Impedance analysis of the half-cell was performed after formation cycle to measure resistance.
- The PE half-cell resistance decreased and then increased again during charging with state of charge (SOC).



Results & Discussion

■ EIS evolution

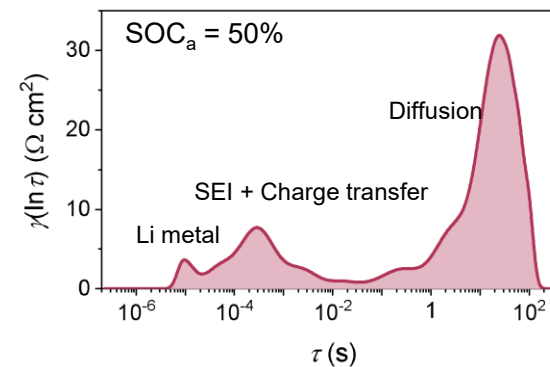
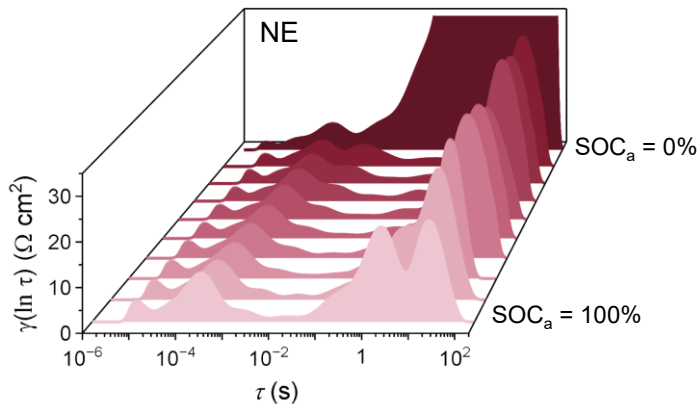
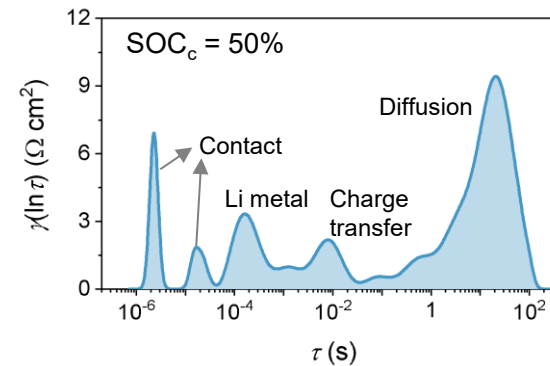
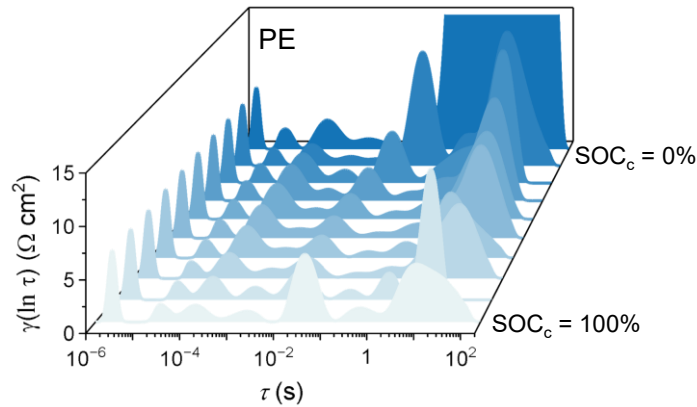
- Similar to the PE half-cell, the NE half-cell resistance decreased and then increased again during charging.
- The before-knee 300cycle showed less difference in resistance with SOC.



Results & Discussion

■ DRT peak assignment

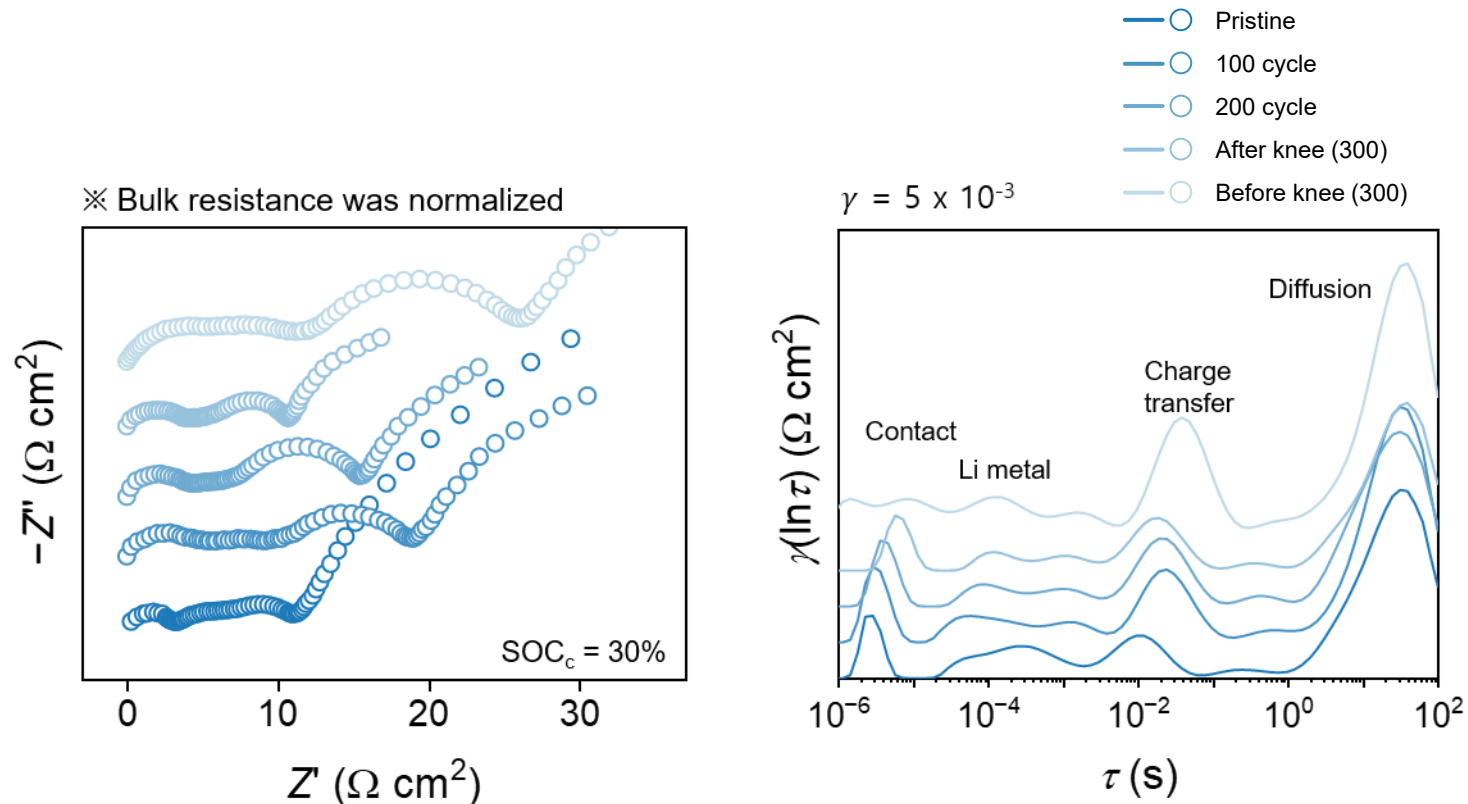
- DRT peak assignment was performed to distinguish the peak signals corresponding to Li metal.



Results & Discussion

■ DRT evolution for PE

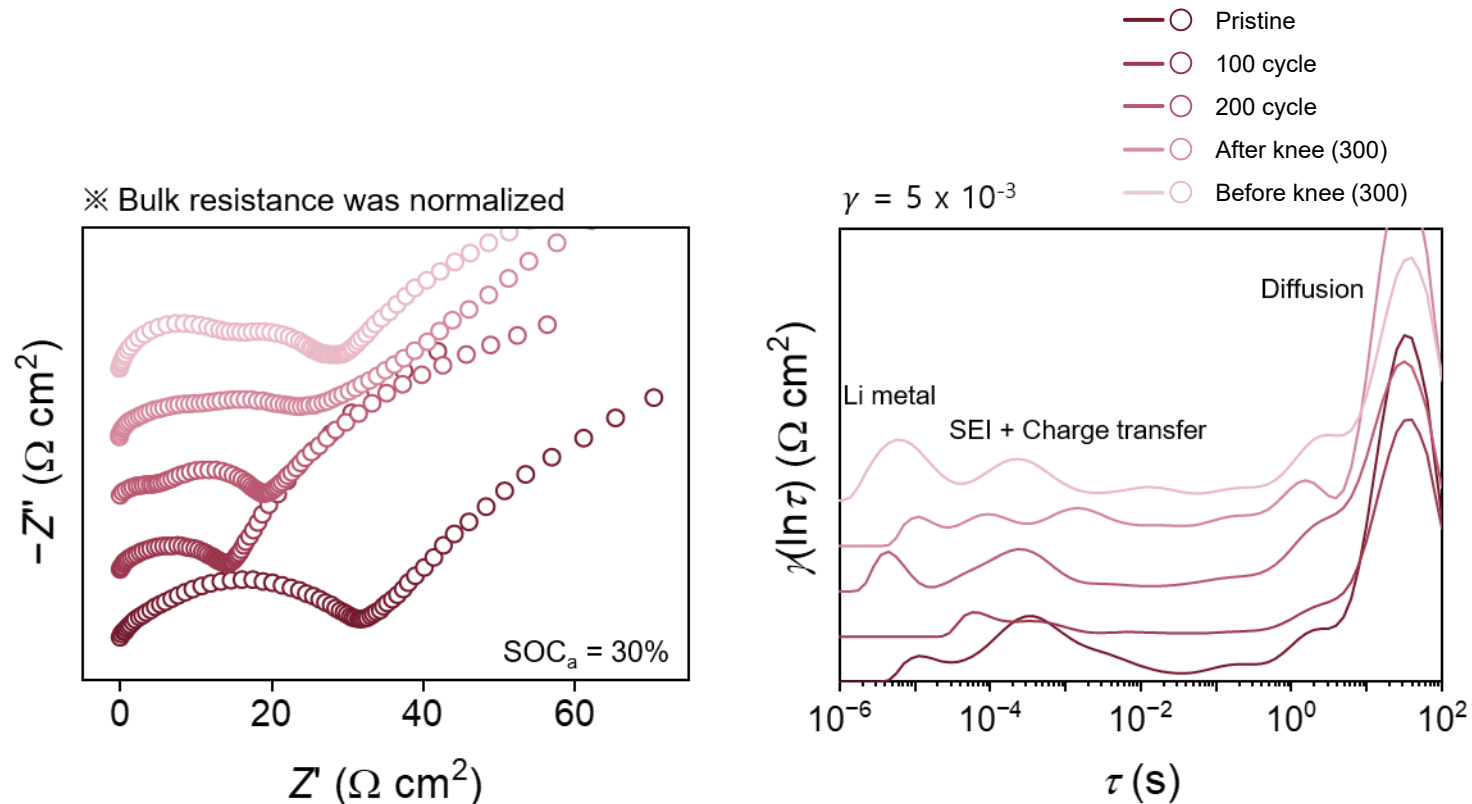
- Charge transfer peak and diffusion peak area increase at before-knee PE.
→ Charge transfer resistance (R_{ct}) and diffusion resistance (R_{diff}) increase



Results & Discussion

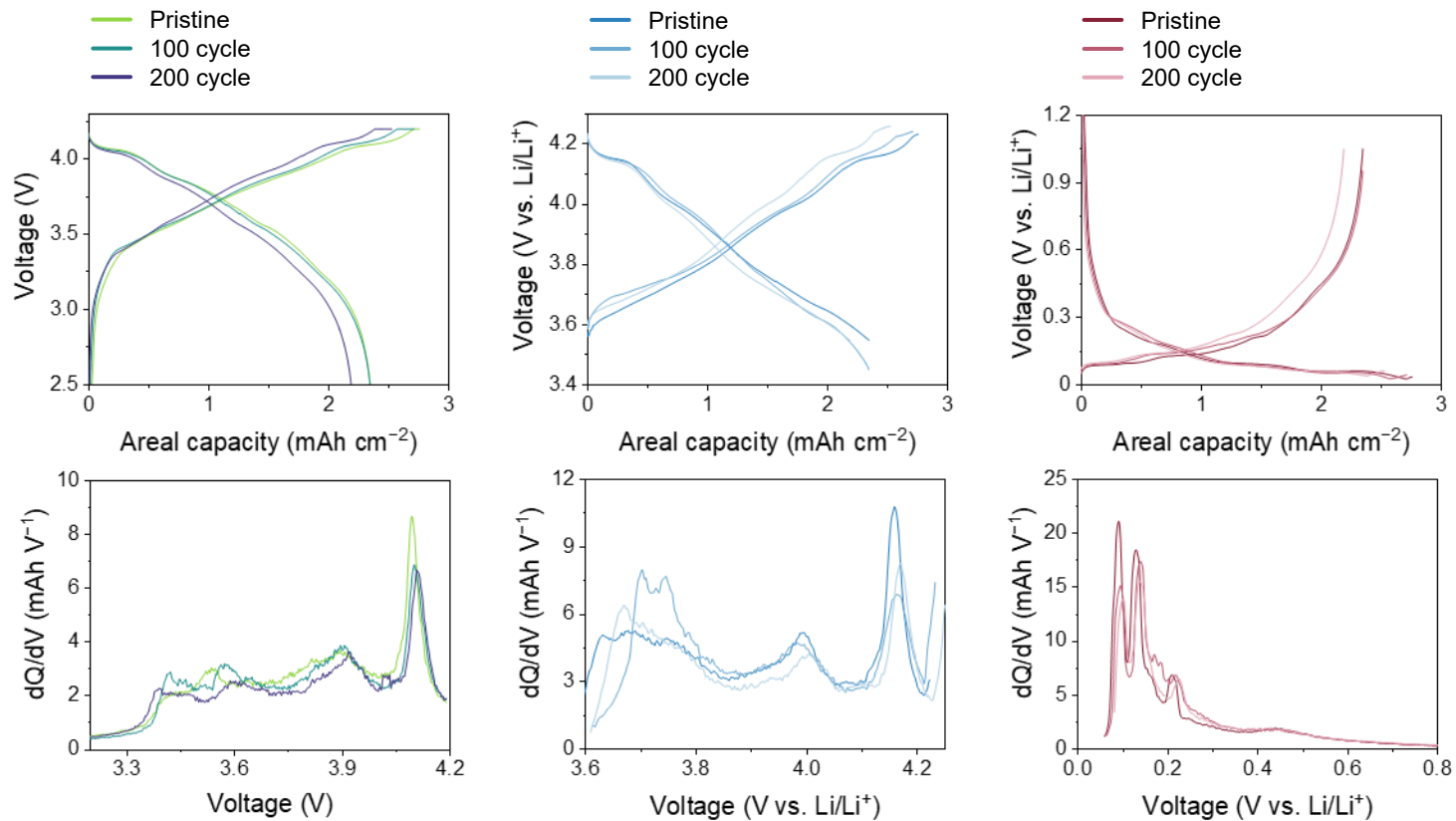
■ DRT evolution for NE

- SEI and charge transfer peak get separated at after-knee NE.
→ Possibility of a difference in R_{ct} between Si and graphite



Results & Discussion

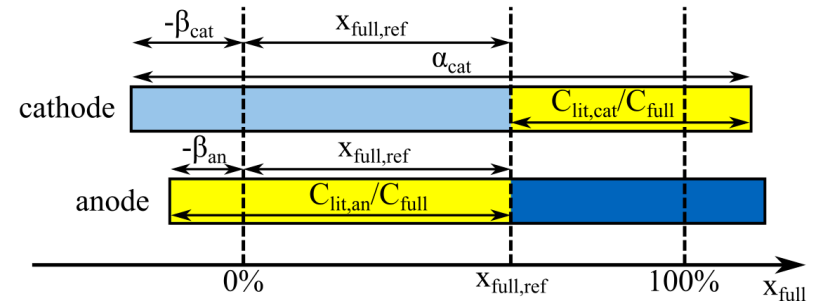
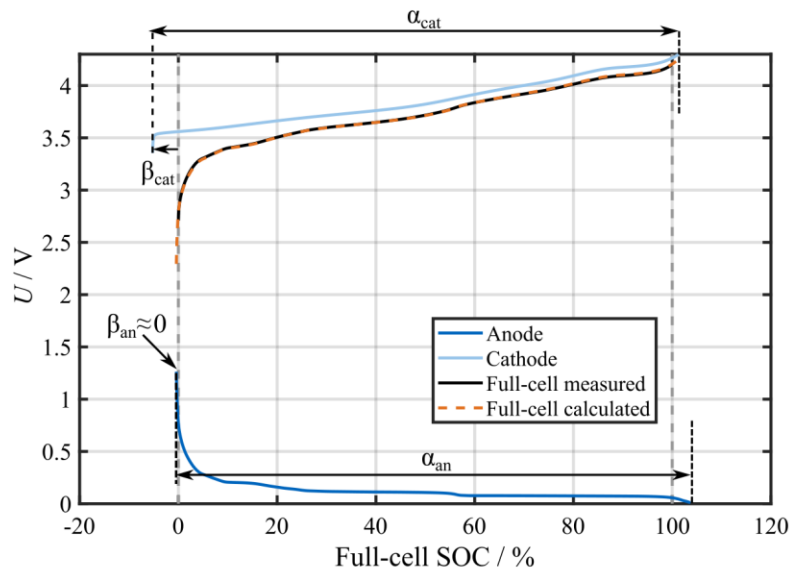
- Three-electrode pouch cell formation
 - Three-electrode pouch cell was formed and converted to ICA curve.
 - Full-cell discharge curves reach the cutoff due to NE up to 200 cycle.



Results & Discussion

■ Degradation mode quantification method

- Degradation modes can be quantified by fitting full-cell and half-cell OCV.
- **Loss of Lithium inventory (LLI)** and **Loss of active material (LAM)** are calculated from the **scaling factor** (α) and **shifting factor** (β) of the half-cell OCV.



$$LLI = 1 - \frac{(\alpha_{PE} + \beta_{PE} - \beta_{NE}) \cdot C_{full}}{(\alpha_{PE,ini} + \beta_{PE,ini} - \beta_{NE,ini}) \cdot C_{full,ini}}$$

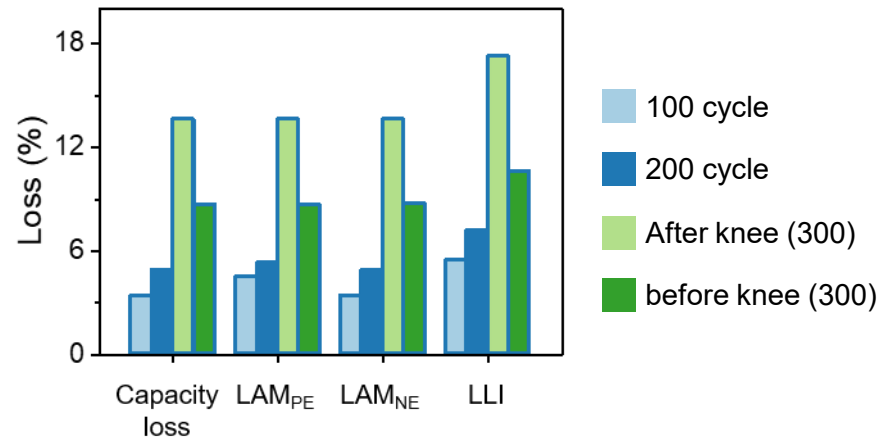
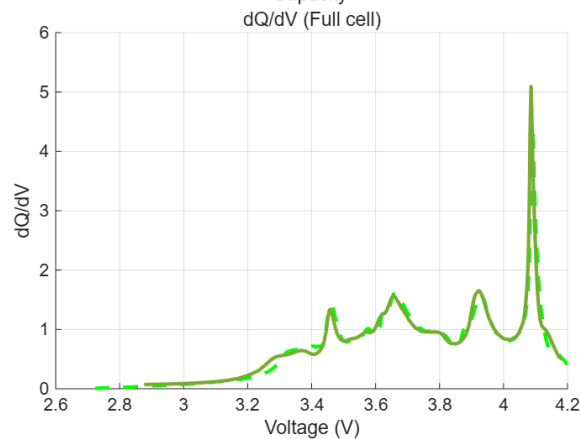
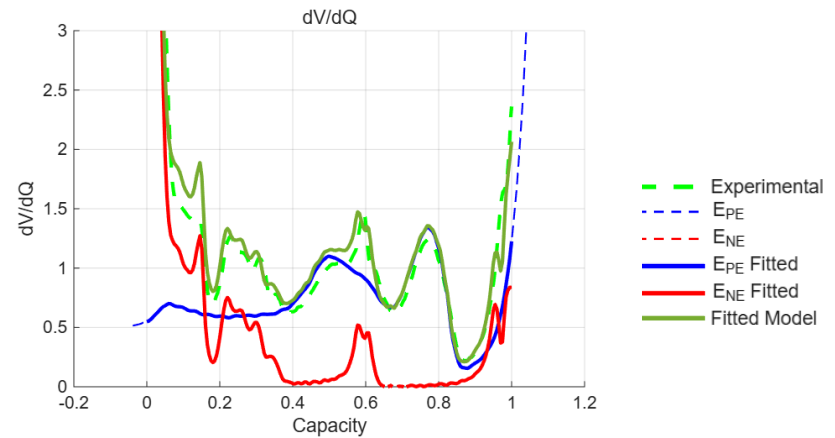
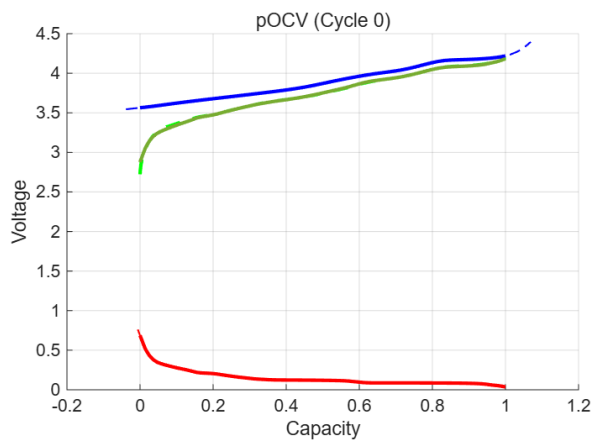
$$LAM_{PE} = 1 - \frac{\alpha_{PE} \cdot C_{full}}{\alpha_{PE,ini} \cdot C_{full,ini}}$$

$$LAM_{NE} = 1 - \frac{\alpha_{NE} \cdot C_{full}}{\alpha_{NE,ini} \cdot C_{full,ini}}$$

Results & Discussion

■ Electrodes degradation mode quantifying

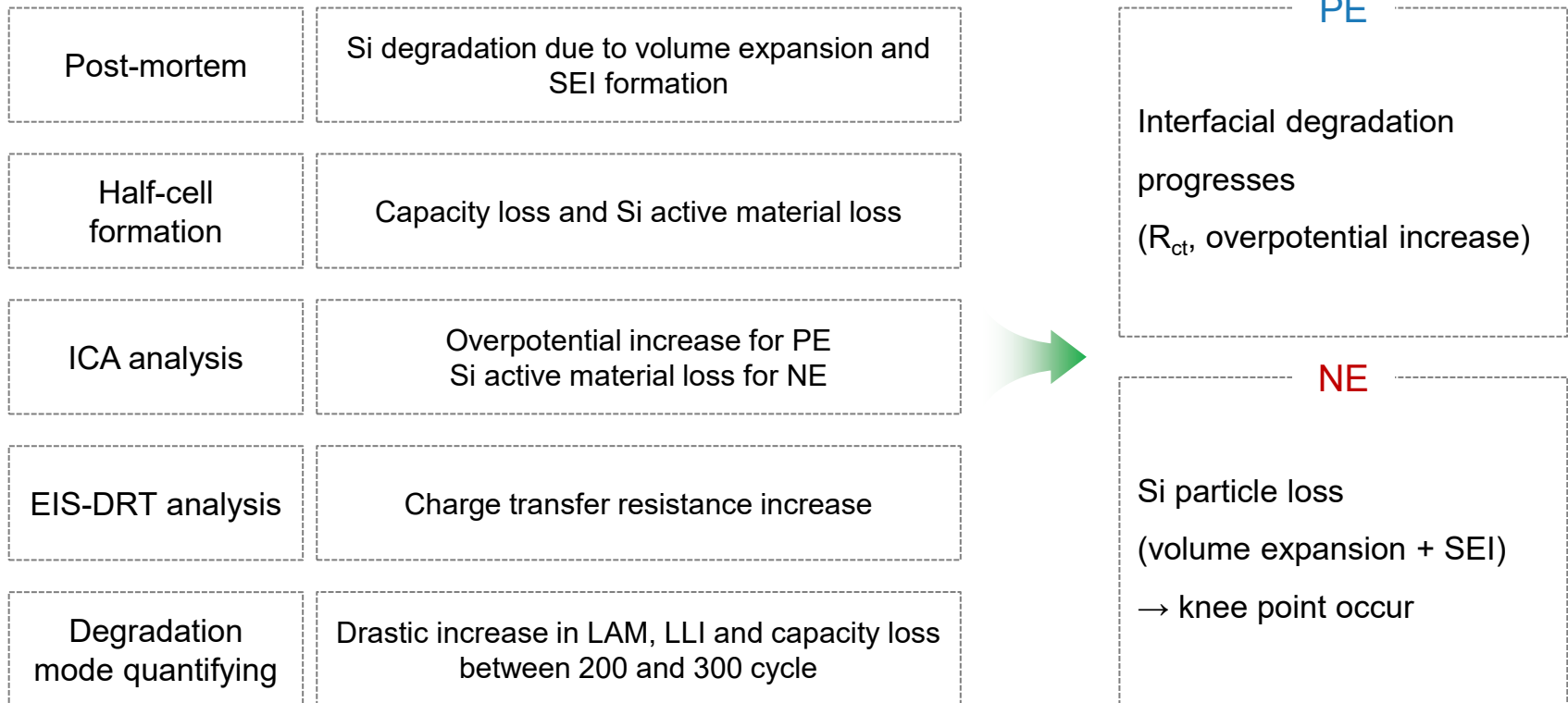
- Degradation mode was quantified via MATLAB code minimizing RMSE.
- Provided full-cell data and half-cell OCV were used for fitting.



Results & Discussion

■ Summary of analysis results

- Further analysis is needed for the electrode at 300 cycle.



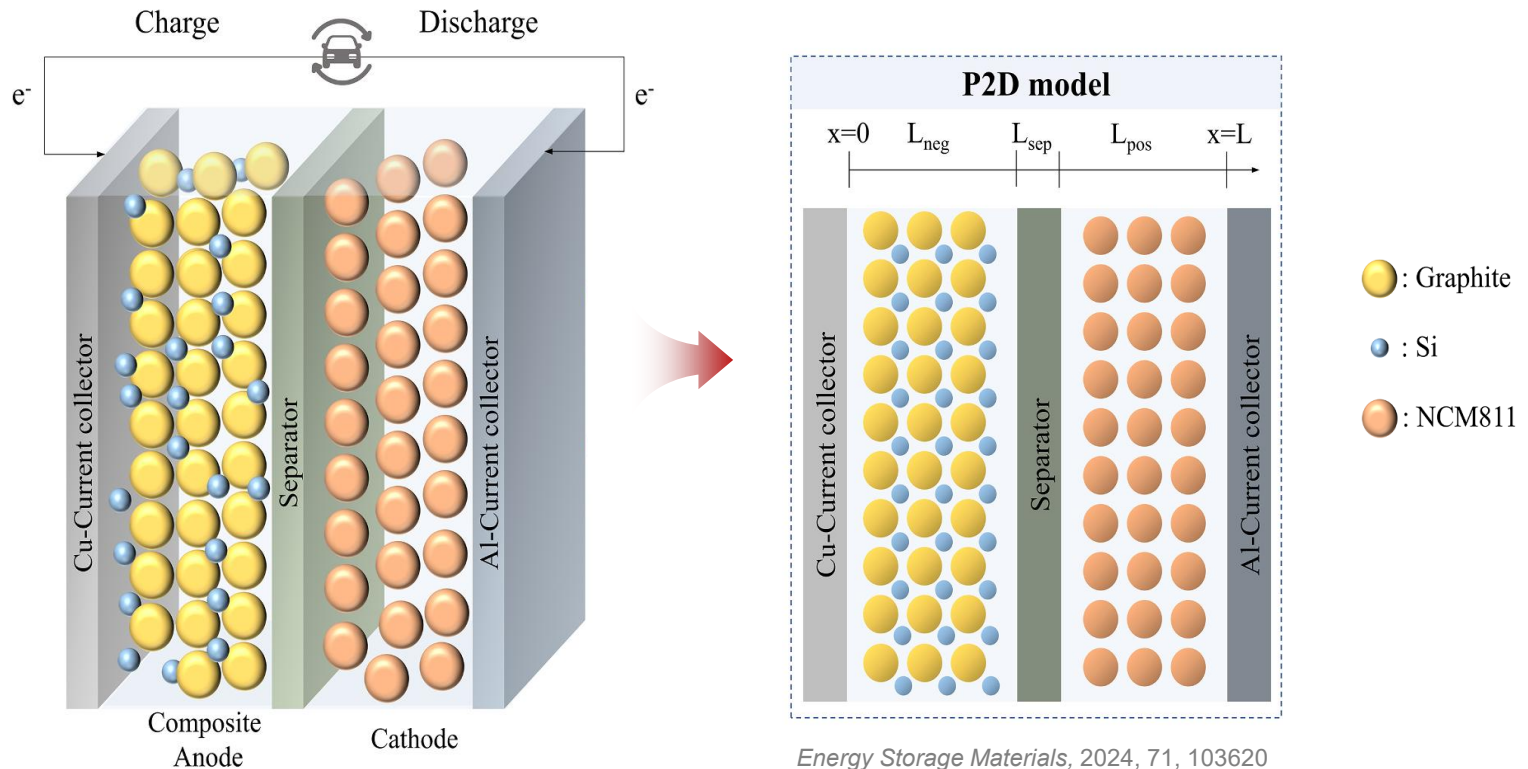
Part 2

Degradation modeling and
degradation mode quantifying for Si-based anodes

Introduction

■ P2D modeling for NCM || Si-graphite cell

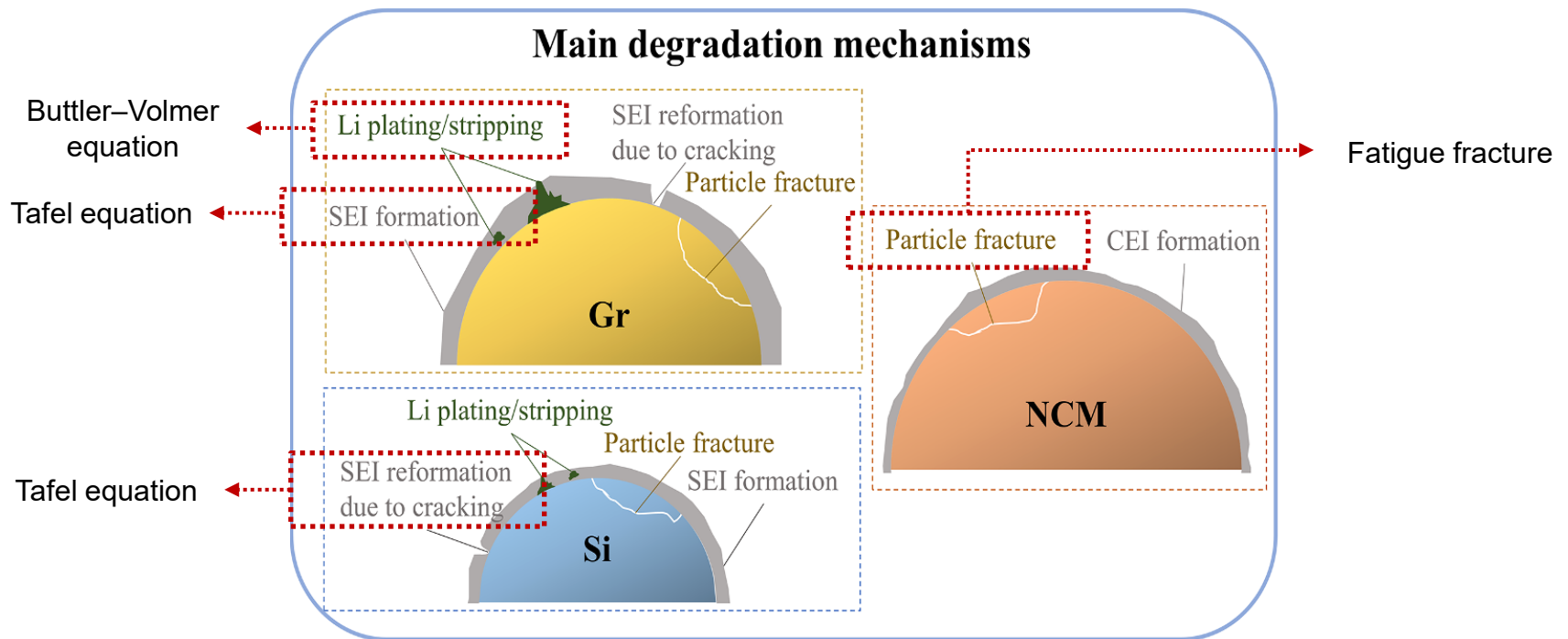
- Modeling helps to identify internal degradation mechanisms during the cycling process.
- P2D modeling was performed to simulate degradation at the electrode particle level.



Introduction

■ Main equations for degradation mechanism

- Degradation equations were applied to COMSOL and simulated for 1C/2C charge, 1C discharge.
- SEI formation, Li plating, and particle fracture were calculated for each particle, from which LLI and LAM were derived.

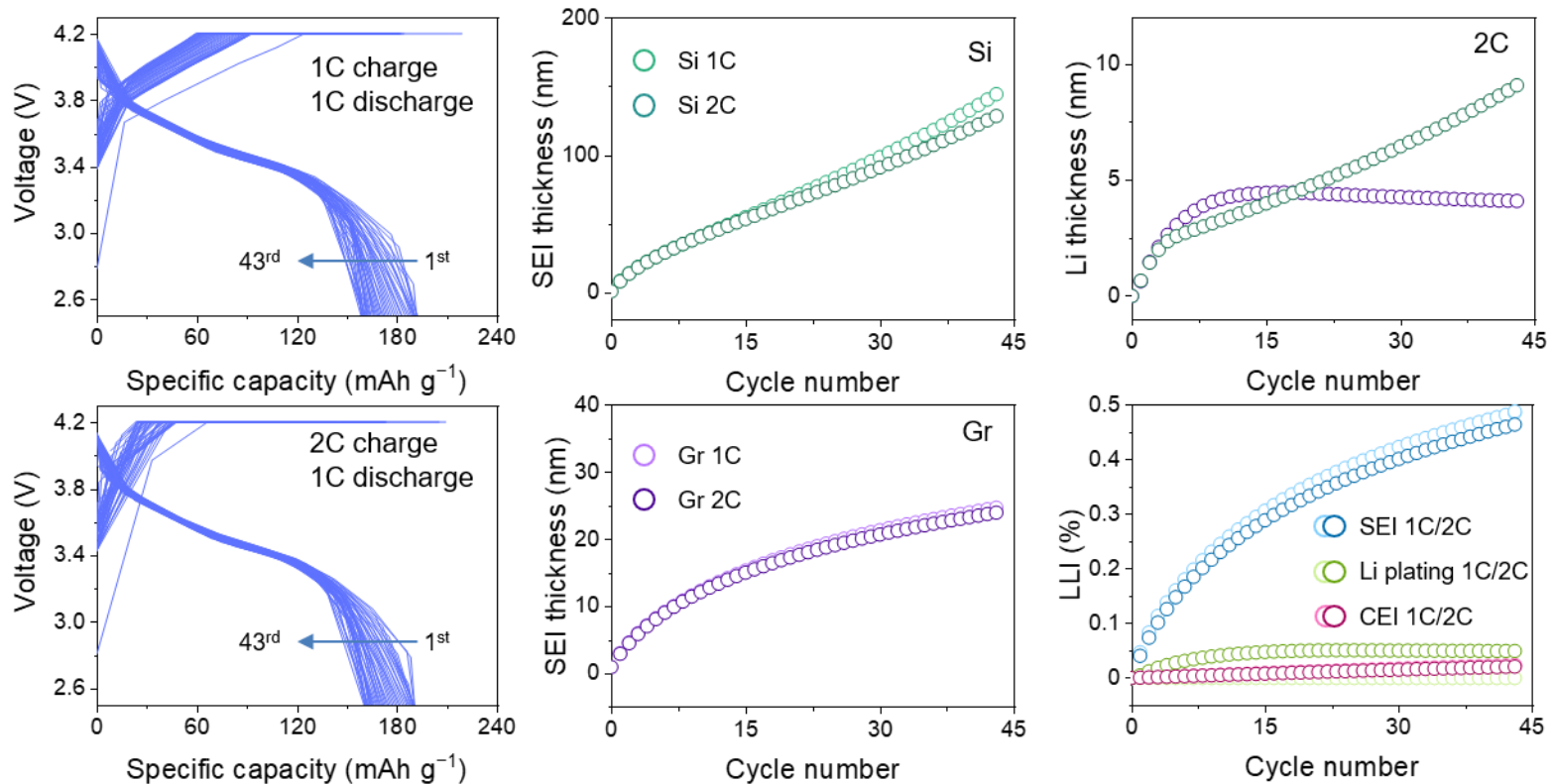


Energy Storage Materials, 2024, 71, 103620

Results & Discussion

■ SEI formation and Li plating for different C-rate

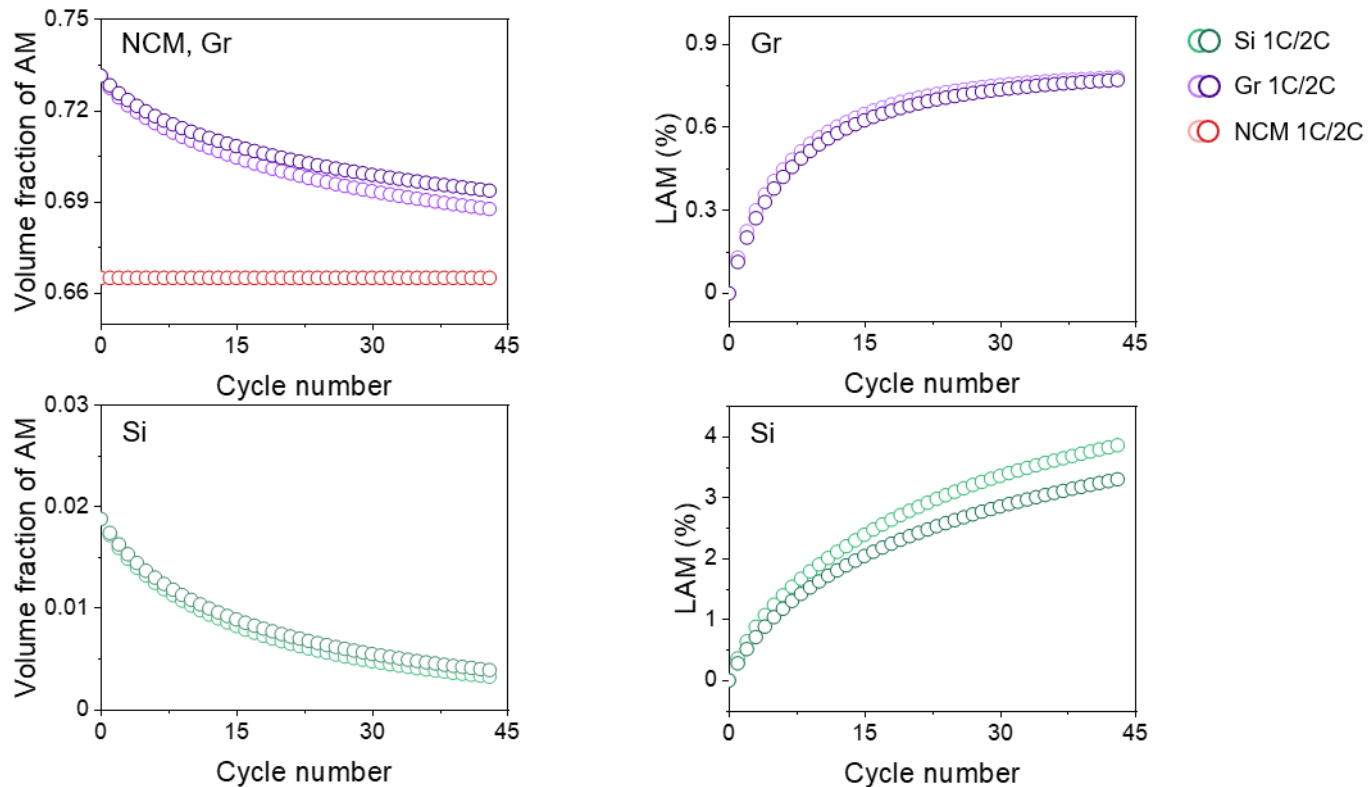
- The data collection resolution during charge/discharge was low due to the increased computational load from the addition of numerous equations.
- SEI growth was more dominant than Li plating in LLI.



Results & Discussion

■ Mechanical degradation for different C-rate

- Si and graphite showed degradation due to a decrease in volume fraction, whereas no such degradation occurred in NCM.
- Active material loss was greater under the 1C charge than under the 2C charge.



Results & Discussion

■ PyBaMM

- PyBaMM (Python Battery Mathematical Modeling) is an open-source battery simulation package written in Python.



```
1 # =====
2 #
3 #
4 #
5 #
6 #
7 #
8 #
9 # Parameter Sweep 설정 파일
10 # 단일 시뮬레이션은 config.py만 사용하고,
11 # 여러 조건을 한 번에 돌리려면 이 파일을 설정하세요.
12 # =====
13
14 #
15 # [1] Sweep 활성화
16 # =====
17
18 SWEEP_ENABLED = True      # True: sweep 실행 / False: config.py 단일 실행
19
20 #
21 # [2] Sweep 모드 선택
22 # "c_rate" : 충/방전 C-rate sweep
```

Advantages

- Fast calculation
- Easy to model with the help of AI
- Existence of sample models and tutorials
- Flexible switching between models (SPM, SPM_e, DFN) and submodels (SEI, Li plating)

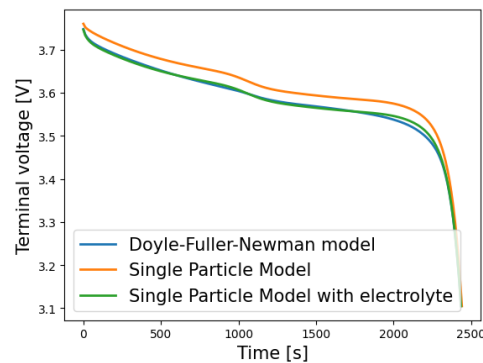
Disadvantages

- Limited geometric flexibility
- Offers less intuitive mesh and solver tuning
- Non-electrochemical physics rely on third-party coupling or approximations

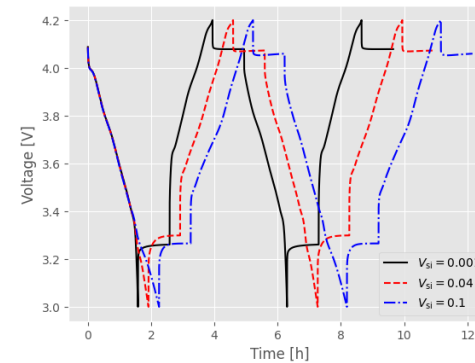
Results & Discussion

■ Advantages of PyBaMM

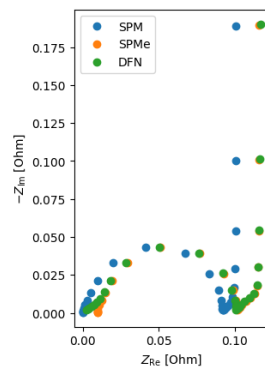
- PyBaMM website contains a collection of example models that demonstrate how to use.



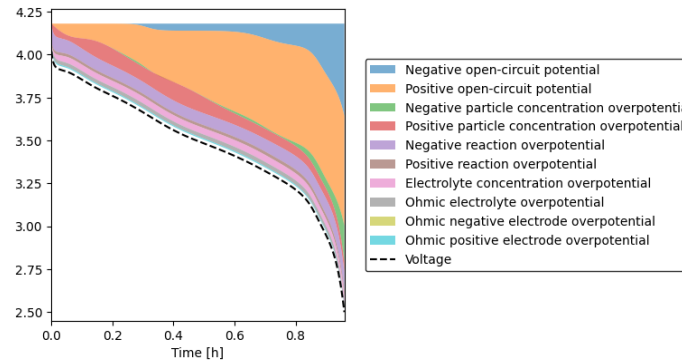
Compare lithium-ion battery models



A composite electrode particle model



EIS simulation

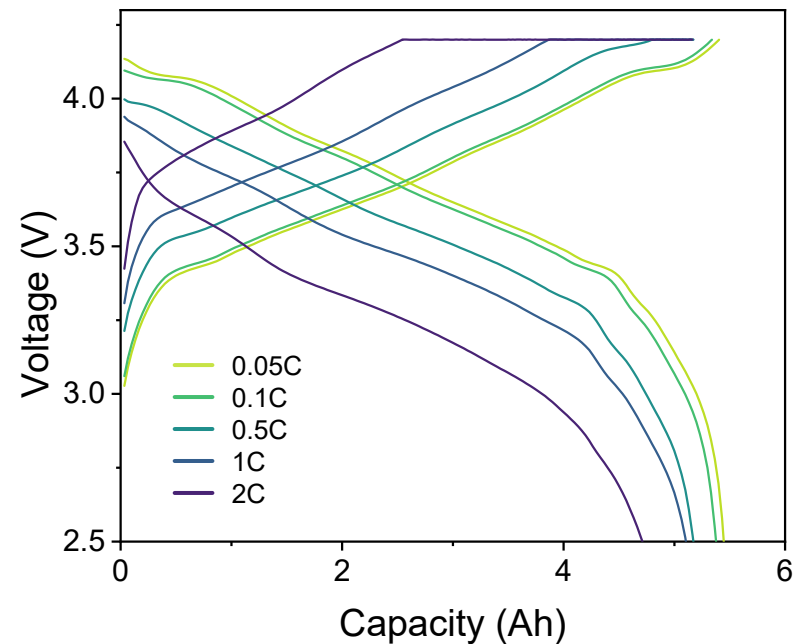
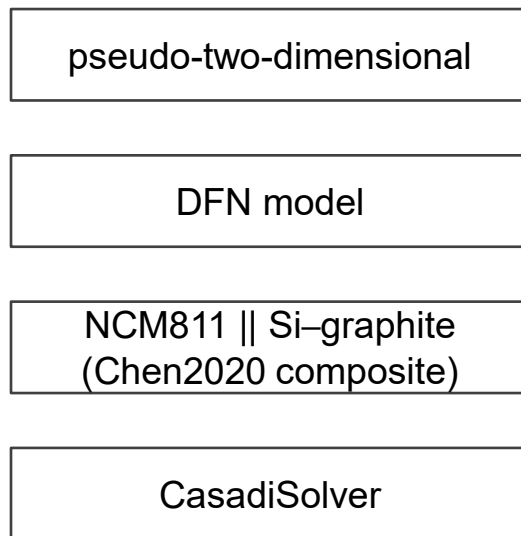


Accounting for formation or storage loss

Results & Discussion

■ Advantages of PyBaMM

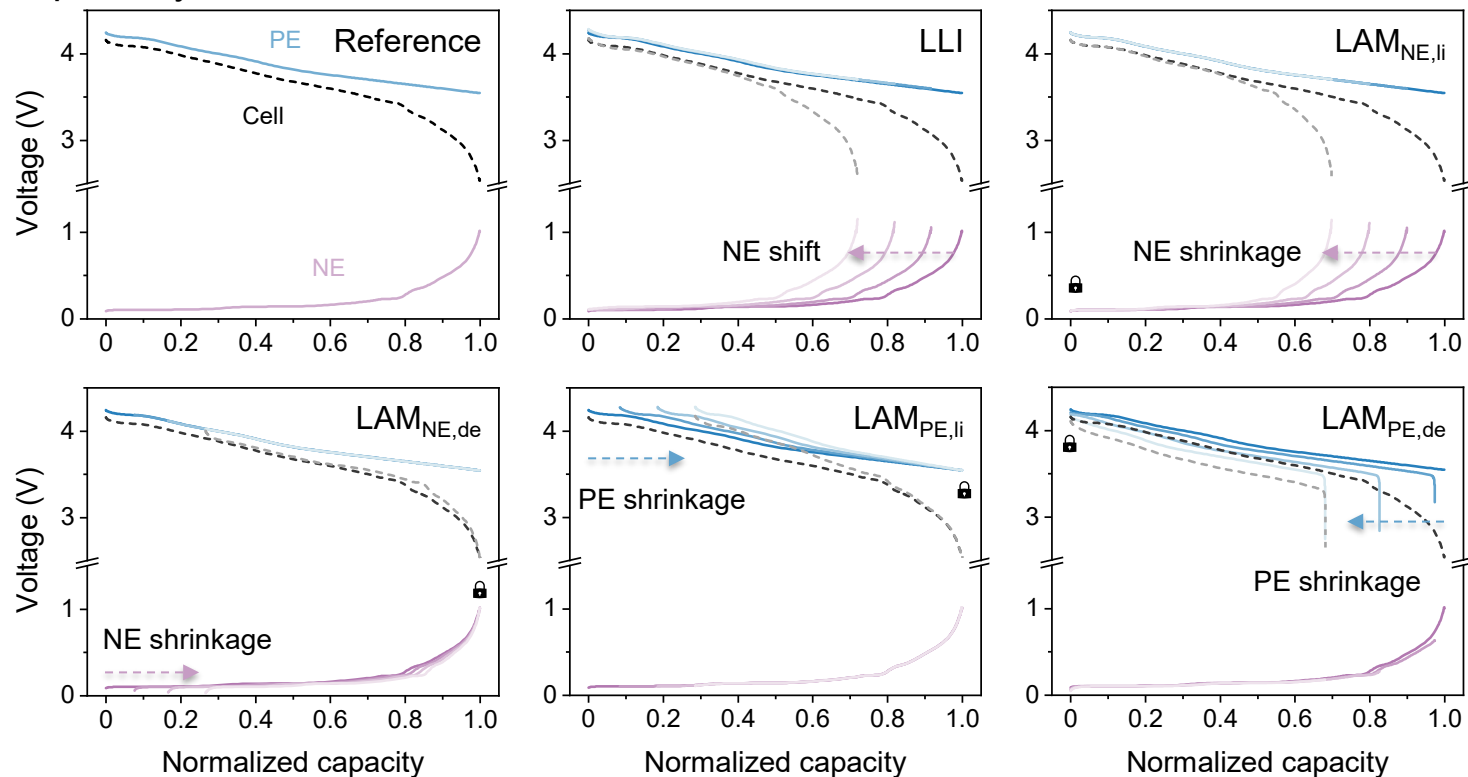
- Unlike COMSOL, PyBaMM allows for flexible coding with the help of AI.
- Parameter sweep code was written by AI without writing the code directly.



Results & Discussion

■ Application of PyBaMM

- Scheme for degradation mode quantifying was constructed using PyBaMM parameter sweep.
- LLI and LAM were implemented by adjusting the Li concentration, volume fraction and porosity of the active material.



Results & Discussion

■ Degradation mode quantifying code

- Get transferred the first version of fitting code

→ Updated the code to increase quantifying quality

```
clear all
clc

% 변수 정의
ocv_path = 'half cell ocv/';
ocv_filename = 'LG_251222_charge';
%ocv_filename = '370mAh_cylindrical_cell_half_cell_GITT_ocv.xlsx';

cycle_path = 'experiment/SiGr/';
cycle_filename = 'pocv_0_05C_charge';
cycle = 0;

fitting_method = "dvdq";
plotting = "pocv";
interactive = true;

% [a_PE, b_PE, a_NE, b_NE]
% b upper bound 0, a lower bound 1로 고정
initial = [1.03, -0.1, 1.08, -0.01]; % initial parameters
ub = [1.1, 0, 1.1, 0]; % upper bound
lb = [1.00, -0.3, 1.00, -0.15]; % lower bound

degradation_mode(cycle_path, cycle_filename, ocv_path, ocv_filename, fitting_method, plotting, interactive);
```

+

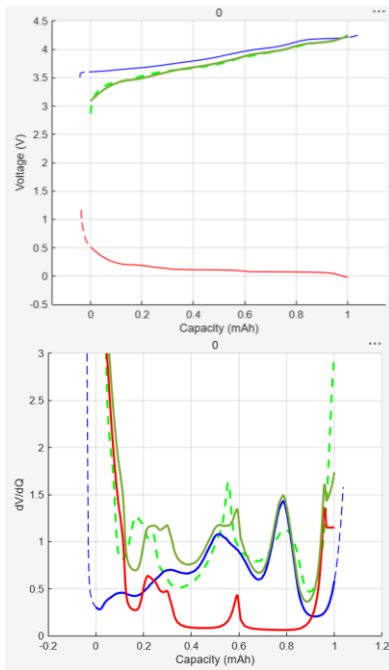
**Updated
function**

1. dQ/dV peak fitting, combined fitting method
2. Peak smoothing control parameter
3. Peak weight factor control parameter

Results & Discussion

- Degradation mode fitting code update
 - dQ/dV fitting method and combined fitting method were added by changing objective function.

pOCV, dV/dQ based

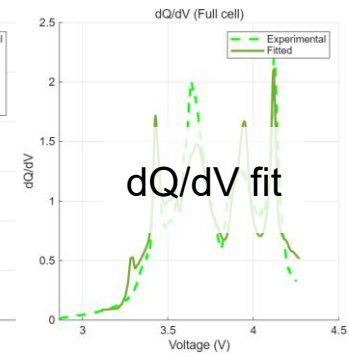
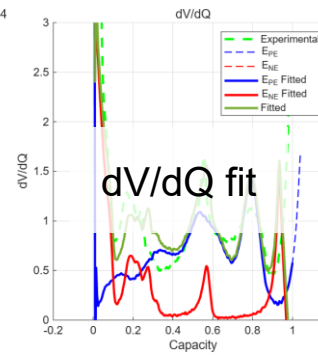
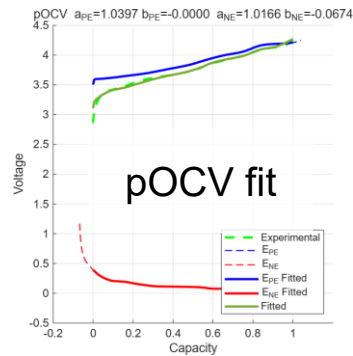


Modified

Objective function

$$J(p) = w_{\text{pocv}} \frac{\text{RMSE}_{\text{pocv}}(p)}{\text{scale}_{\text{pocv}}} + w_{\text{dvdq}} \frac{\text{RMSE}_{\text{dvdq}}(p)}{\text{scale}_{\text{dvdq}}} + w_{\text{dqdv}} \frac{\text{RMSE}_{\text{dqdv}}^w(p)}{\text{scale}_{\text{dqdv}}}$$

$$\hat{p} = \underset{lb \leq p \leq ub}{\text{argmin}} J(p)$$



Discussion

■ Future plans

- Post-mortem analysis
→ 300 cycle electrodes (XPS, XRD, cross-sectional SEM)
- Collecting reliable data for before-knee half-cell formation and GITT
- EIS-DRT analysis to ensure consistent tendency with cycling progression
- Additional collection of three-electrode pouch cell data

■ Discussion points

- Improved fitting algorithm for advancing degradation mode quantification
- Applications of machine learning in the ICA degradation analysis process

Thank you

For more information or any questions,
please e-mail at ga117@hanyang.ac.kr



■ Degradation reaction equation applied in COMSOL – SEI formation

SEI formation

$$j_{SEI-formation,C_6} = -a_{C_6}j_{0,SEI}\exp\left(-\frac{\alpha_{c,SEI}F}{RT}\eta_{SEI-formation,C_6}\right)\exp(-\lambda_{limit}\delta_{film,C_6})$$

$$j_{SEI-formation,Si} = -a_{Si}j_{0,SEI}\exp\left(-\frac{\alpha_{c,SEI}F}{RT}\eta_{SEI-formation,Si}\right)\exp(-\lambda_{limit}\delta_{film,Si})$$

$$\eta_{SEI-formation,C_6} = \phi_{s,C_6} - \phi_e - U_{SEI} - \frac{j_{tot,C_6}R_{film,C_6}}{a_{C_6}}$$

$$\eta_{SEI-formation,Si} = \phi_{s,Si} - \phi_e - U_{SEI} - \frac{j_{tot,Si}R_{film,Si}}{a_{Si}}$$

SEI reformation
due to cracking

$$j_{SEI-crack,C_6} = -a_{C_6}j_{0,SEI}f_{C_6,crack}(x)\exp\left(-\frac{\alpha_{c,SEI}F}{RT}\eta_{SEI-crack,C_6}\right)\exp(-\lambda_{limit,crack}\delta_{film,C_6})$$

$$j_{SEI-crack,Si} = -a_{Si}j_{0,SEI}f_{Si,crack}(y)\exp\left(-\frac{\alpha_{c,SEI}F}{RT}\eta_{SEI-crack,Si}\right)\exp(-\lambda_{limit,crack}\delta_{film,Si})$$

$$\eta_{SEI-crack,C_6} = \phi_{s,C_6} - \phi_e - U_{SEI}$$

$$\eta_{SEI-crack,Si} = \phi_{s,Si} - \phi_e - U_{SEI}$$

Lithium balancing
and LLI

$$\frac{\partial c_{SEI-tot,C_6}}{\partial t} = \frac{\partial c_{SEI-formation,C_6}}{\partial t} + \frac{\partial c_{SEI-crack,C_6}}{\partial t} = -\frac{j_{SEI-formation,C_6} + j_{SEI-crack,C_6}}{2F}$$

$$\frac{\partial c_{SEI-tot,Si}}{\partial t} = \frac{\partial c_{SEI-formation,Si}}{\partial t} + \frac{\partial c_{SEI-crack,Si}}{\partial t} = -\frac{j_{SEI-formation,Si} + j_{SEI-crack,Si}}{2F}$$

$$Q_{SEI}^{Li^+} = 2A_{electrode} \int_{L_{neg}} (c_{SEI-tot,C_6} + c_{SEI-tot,Si}) F dl$$

■ Degradation reaction equation applied in COMSOL – Li plating/stripping

Li plating

$$j_{LP,C_6} = a_{C_6} F k_{LP} \left(c_{Li,C_6} \exp\left(\frac{\alpha_{a,Li} F}{RT} \eta_{Li,C_6}\right) - c_e \exp\left(\frac{\alpha_{c,Li} F}{RT} \eta_{Li,C_6}\right) \right), \eta_{Li,C_6} < 0$$

$$j_{LP,Si} = a_{Si} F k_{LP} \left(c_{Li,Si} \exp\left(\frac{\alpha_{a,Li} F}{RT} \eta_{Li,Si}\right) - c_e \exp\left(\frac{\alpha_{c,Li} F}{RT} \eta_{Li,Si}\right) \right), \eta_{Li,Si} < 0$$

$$\eta_{Li,C_6} = \phi_{s,C_6} - \phi_e - \frac{j_{tot,C_6}}{a_{C_6}} R_{film,C_6}$$

$$\eta_{Li,Si} = \phi_{s,Si} - \phi_e - \frac{j_{tot,Si}}{a_{Si}} R_{film,Si}$$

Li stripping

$$j_{SR,C_6} = a_{C_6} F k_{LP} \frac{\psi_{rev}}{\delta_{film,C_6}} \left(c_{EC} \exp\left(\frac{\alpha_{a,Li} F}{RT} \eta_{Li,C_6}\right) - c_{Li,C_6} \exp\left(\frac{\alpha_{c,Li} F}{RT} \eta_{Li,C_6}\right) \right),$$

$$\eta_{Li,C_6} \geq 0, c_{Li,C_6} > 0$$

$$j_{SR,Si} = a_{Si} F k_{LP} \frac{\psi_{rev}}{\delta_{film,Si}} \left(c_{EC} \exp\left(\frac{\alpha_{a,Li} F}{RT} \eta_{Li,Si}\right) - c_{Li,Si} \exp\left(\frac{\alpha_{c,Li} F}{RT} \eta_{Li,Si}\right) \right),$$

$$\eta_{Li,Si} \geq 0, c_{Li,Si} > 0$$

Lithium balancing
and LLI

$$\frac{\partial c_{Li,C_6}}{\partial t} = -\frac{j_{LP,C_6} + j_{SR,C_6}}{F}, \frac{\partial c_{Li,Si}}{\partial t} = -\frac{j_{LP,Si} + j_{SR,Si}}{F}$$

$$Q_{Li}^{Li^+} = A_{electrode} \int_{L_{neg}} (c_{Li,C_6} + c_{Li,Si}) F dl$$

Appendix

■ Degradation reaction equation applied in COMSOL – mechanical (1)

Radial stress	$\sigma_r(r) = \frac{2Y\Omega}{9(1-\nu)} [C_{av}(R) - C_{av}(r)]$
Tangential stress	$\sigma_\theta(r) = \frac{2Y\Omega}{9(1-\nu)} [2C_{av}(R) + C_{av}(r) - 3C(r)]$
Elastic strain energy	$e(r) = \frac{\sigma_r^2(r) + \sigma_\theta^2(r) - 2\sigma_r(r)[2\sigma_r(r) + \sigma_\theta(r)]}{2Y}$
Total elastic strain energy	$E_T = \int_0^{L_{anode/cathode}} \left(4\pi \int_0^R e(r)r^2 dr \right) dl$
Von Mises stress	$\sigma_{vonmises}(r) = \sigma_\theta(r) - \sigma_r(r) $

Appendix

■ Degradation reaction equation applied in COMSOL – mechanical (2)

Accumulation of von Mises stresses
in the active electrode material

$$I_{fatigue} = \varepsilon_s \int_0^{t_{end}} \int_0^{L_{anode/cathode}} \left(4\pi \int_0^R \sigma_{vonmises}(r) r^2 dr \right) dl dt$$

Electrode fatigue factor

$$\zeta_{fatigue} = \exp(-\omega_{fatigue} I_{fatigue})$$

Limiting total elastic strain,
Young's modulus

$$E_{T,lim} = \zeta_{fatigue} E_{T,lim,0}, Y = \zeta_{fatigue} Y_0$$

Ratio of the current extra strain
energy to the average extra strain

$$d_{LAM} = \max\left(0, \frac{E_T - E_{T,lim}}{E_{T,sn} - E_{T,lim}}\right) = \max\left(0, \frac{E_T/E_{T,sn} - \beta\zeta_{fatigue}}{1 - \beta\zeta_{fatigue}}\right)$$

Decrease of volume fraction

$$\frac{\partial \varepsilon_s}{\partial t} = -d_{LAM} k_{LAM} \varepsilon_s$$

Li balancing

$$\frac{\partial Q_{LAM}^{Li^+}}{\partial t} = \xi_{SOL} V_s C_{s,max} F \frac{\partial \varepsilon_s}{\partial t}$$

Appendix

■ Degradation model results from the reference

